



K. J. Somaiya College of Engineering, Mumbai-77
(A Constituent College of Somaiya Vidyavihar University)
Department of Computer Engineering

Batch: _____ **Roll No.:** _____
Experiment No. _____
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

Title: Implementation of Knapsack Problem using Greedy strategy

Objective: To learn the Greedy strategy of solving the problems for different types of problems

CO to be achieved:

CO 2 Describe various algorithm design strategies to solve different problems and analyse Complexity.

Books/ Journals/ Websites referred:

1. Ellis horowitz, Sarataj Sahni, S.Rajsekaran," Fundamentals of computer algorithm", University Press
2. T.H.Cormen ,C.E.Leiserson,R.L.Rivest and C.Stein," Introduction to algortihtms",2nd Edition ,MIT press/McGraw Hill,2001
3. <http://lcm.csa.iisc.ernet.in/dsa/node184.htm>
4. <http://students.ceid.upatras.gr/~papagel/project/kruskal.htm>
- 5.

<http://www.personal.kent.edu/~rmuhamma/Algorithms/MyAlgorithms/Grap hAlgor/kruskalAlgor.html>

6. <http://lcm.csa.iisc.ernet.in/dsa/node183.html>
7. <http://students.ceid.upatras.gr/~papagel/project/prim.htm>
8. <http://www.cse.ust.hk/~dekai/271/notes/L07/L07.pdf>

Pre Lab/ Prior Concepts: _____

Data structures, Concepts of algorithm analysis

Historical Profile:



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The knapsack problem represents constraint satisfaction optimization problems' family. Based on nature of constraints, the knapsack problem can be solved with various problem solving strategies. Typically, these problems represent resource optimization solution.

Given a set of n inputs. · Find a subset, called feasible solution, of the n inputs subject to some constraints, and satisfying a given objective function. · If the objective function is maximized or minimized, the feasible solution is optimal. · It is a locally optimal method.

New Concepts to be learned:

Application of algorithmic design strategy to any problem, Greedy method of problem solving Vs other methods of problem solving, optimality of the solution, knapsack problem and their applications

Knapsack Problem Algorithm

Algorithm GreedyKnapsack (m, n)

```
// P[1 : n] and w[1 : n] contain the profits and weights respectively of
// Objects ordered so that  $p[i] / w[i] > p[i + 1] / w[i + 1]$ .
// m is the knapsack size and x[1: n] is the solution vector.
{
    for i := 1 to n do x[i] := 0.0           // initialize x
    U := m;
    for i := 1 to n do
    {
        if (w(i) > U) then break;
        x [i] := 1.0; U := U - w[i];
    }
    if (i ≤ n) then x[i] := U / w[i];
}
```

Example: Knapsack Problem



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Analysis of Knapsack Problem algorithm:

Conclusion: